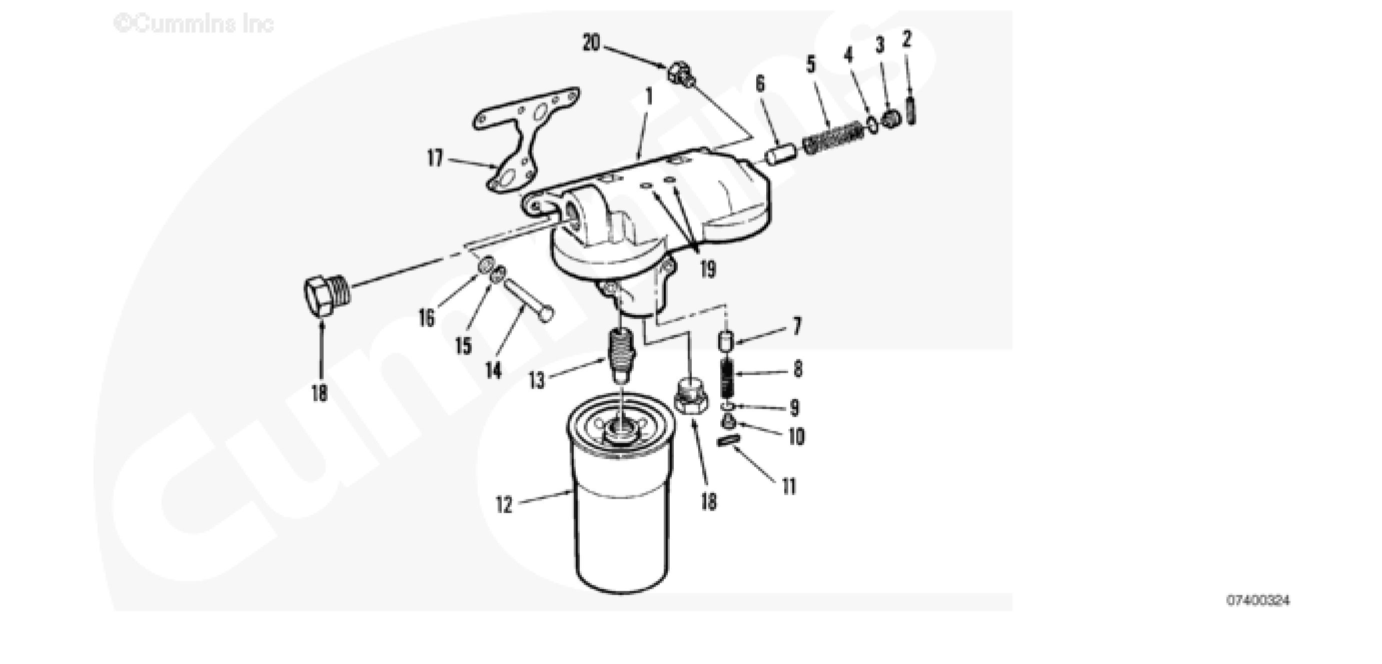
Trainings

Exploded View



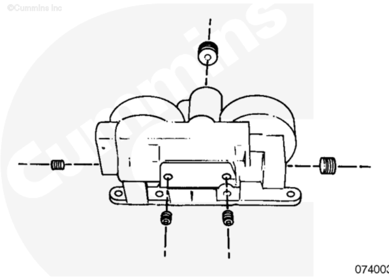
1. Lubricating oil filter head
2. Roll pin
3. Stop pressure regulator
4. O-ring
5. Bypass valve spring
6. Bypass valve plunger
7. Piston cooling valve plunger
8. Piston cooling valve spring
9. O-ring
10. Stop pressure regulator
11. Roll pin
12. Lubricating oil filter element
13. Oil filter adapter
14. Capscrew
15. Lock washer
16. Plain washer
17. Lubricating oil filter head gasket
18. Pipe plug, 1-in
19. Pipe plug, 1/8-in
20. Pipe plug, 3/8-in.

General Information

The oil filter head contains two spring loaded plungers. One plunger controls the oil pressure for the piston cooling nozzles. The second spring plunger will bypass oil if a filter element becomes plugged or clogged.

When installing a new filter element, **always** check to be sure there is **not** interference between the filter head and the element.

Three different designs of filter adapters have been used in the filter head. The revised design has slots machined on the inside diameter. This allows any ¾-in drive tool to be used to remove the adapter. It is **not** necessary to remove the adapter unless it is damaged.



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Preparatory Steps

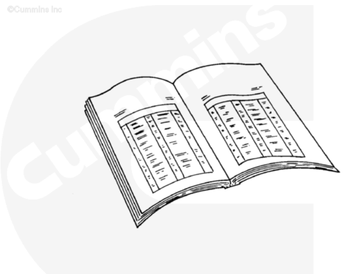
⚠ **WARNING** ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ **WARNING** ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Remove the oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html).



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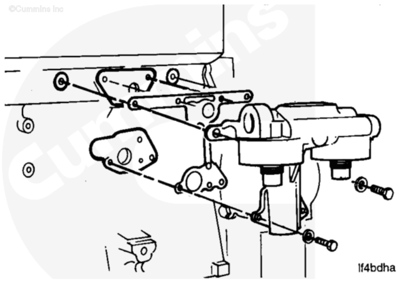
Remove

Remove the six oil filter head mounting capscrews.

Remove the oil filter head assembly.

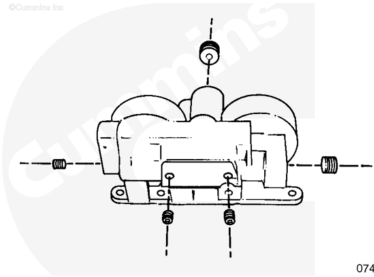
Remove and discard the gasket.



Disassemble

Note : Older filter heads have pipe plugs. Newer filter heads have straight-thread o-ring plugs.

Remove the plugs.

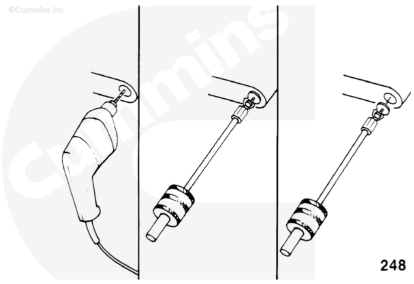


Use a drill, a sheet metal screw, and the below listed parts from the Light Duty Puller Kit, Part Number 3375784.

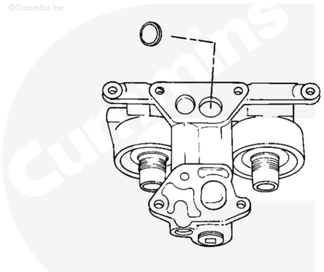
- Slide hammer
- Hook.

Drill a hole in the cup plug and install a sheet metal screw.

Attach the slide hammer to the sheet metal screw.



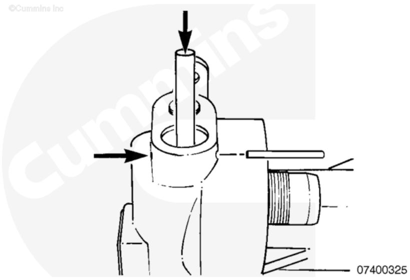
Remove the cup plug.



⚠ WARNING ⚠

To reduce the possibility of personal injury, hold the regulator stop in position when removing the roll pin. The regulator stop is under pressure and can pop out.

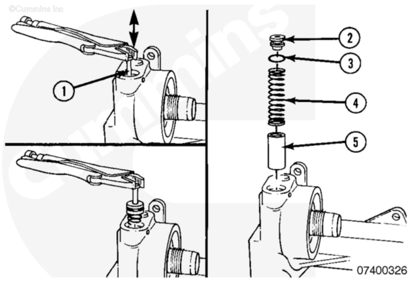
Remove the roll pin.



A ¼-20 UNC-2A capscrew (1) can be used to remove the regulator stop (2).

Remove the o-ring (3), bypass valve spring (4), and bypass valve regulator plunger (5).

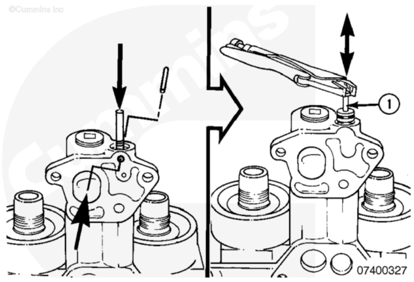
Discard the o-ring.



⚠ WARNING ⚠

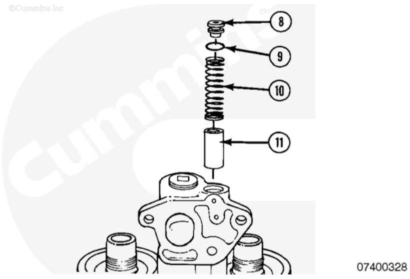
To reduce the possibility of personal injury, hold the regulator stop in position when removing the roll pin. The regulator stop is under pressure and can pop out.

A ¼-20 UNC-2A capscrew can be used to remove the regulator stop.



Remove the pressure regulator stop (8), o-ring (9), piston cooling spring (10), and piston cooling valve plunger (11).

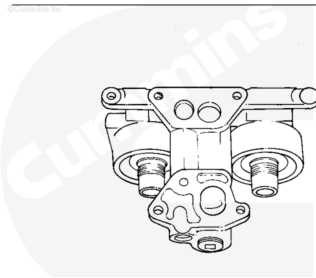
Discard the o-ring.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ WARNING ⚠

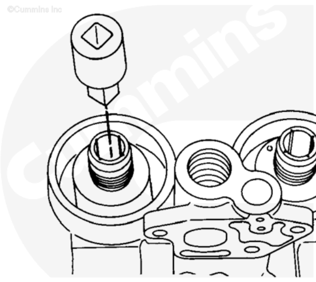
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the lubricating oil filter head and dry with compressed air.

The solvent **must** be removed from the oil drillings.

Inspect the filter head for cracks and damage.

The filter head **must** be replaced if cracked or damaged.



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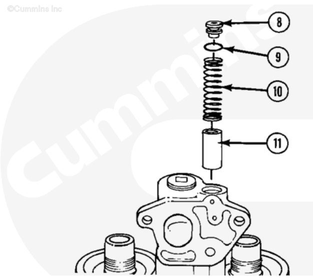
Inspect the lubricating oil filter head for adapter for damage.

If the adapter is damaged, it **must** be replaced. Refer to Procedure 007-018 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-018-tr.html).



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⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

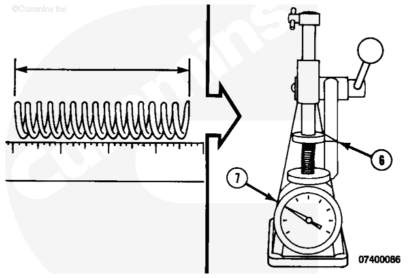
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean lubricating oil filter head parts in solvent and dry with compressed air.

Inspect the parts for damage. If the parts are damaged, they **must** be replaced.

Check the length of the springs.

Spring Free Length	
Piston Cooling	**Bypass**
88.90 mm [3.50 in]	88.90 mm [3.50 in]



Use valve spring tester, Part Number 3375182, or equivalent to measure the spring force at the (7) at the spring working height (6).

Spring Working Height (6)	
Piston Cooling	**Bypass**
50.80 mm [2.000 in]	50.80 mm [2.000 in]

Spring Force (7)			
	n.m		ft-lb
Piston Cooling	49.0	MIN	36.0
	54.0	MAX	40.0
Bypass	49.0	MIN	36.0
	54.0	MAX	40.0

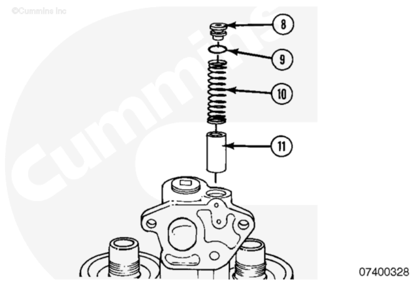
Assemble

Lubricate the parts with clean engine oil.

Install the new o-ring (9) onto the regulator stop (8).

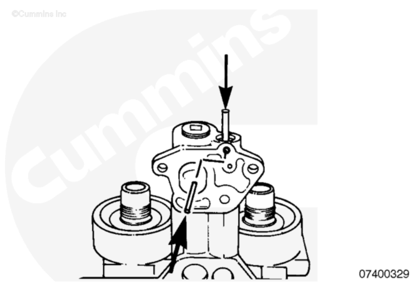
Install the piston cooling valve regulator plunger (11) into the oil filter head.

Install the piston cooling pressure spring (10) and the pressure regulator stop (8) with o-ring (9) into the oil filter head.



⚠ WARNING ⚠

To reduce the possibility of personal injury, hold the regulator stop in position when removing the roll pin. The regulator stop is under pressure and can pop out.



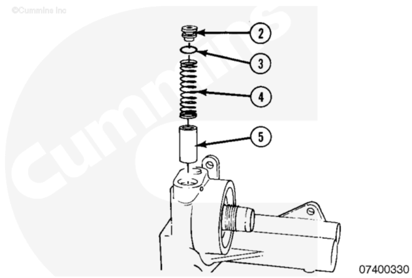
Use a mandrel and an arbor press to hold the regulator stop in position while installing the roll pin.

Lubricate the parts with clean engine oil.

Install the new o-ring (3) onto the regulator stop (2).

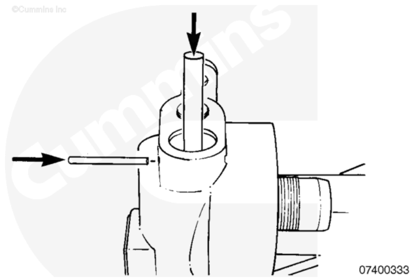
Install the bypass valve regulator plunger (5) into the oil filter head.

Install the bypass valve spring (4) and regulator stop (2) with o-ring (3).



⚠ WARNING ⚠

To reduce the possibility of personal injury, hold the regulator stop in position when removing the roll pin. The regulator stop is under pressure and can pop out.

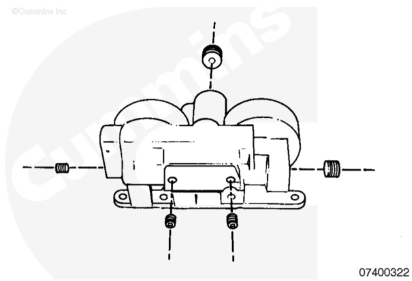


Use a mandrel and an arbor press to hold the regulator stop in position while installing the roll pin.

Apply pipe plug sealant, Part Number 3375066, or teflon tape, or equivalent to the pipe plug threads.

Install the pipe plugs.

Tighten the pipe plugs.



Torque Value:
1-inch pipe plug

1. 75 n•m [55 ft-lb]

Torque Value:
3/8-inch pipe plug

1. 27 n•m [20 ft-lb]

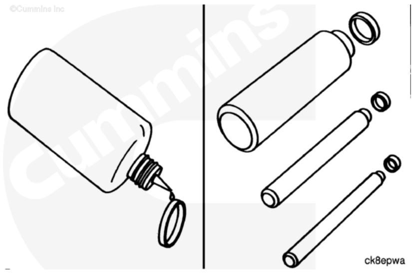
Torque Value:
1/8-inch pipe plug

1. 14 n•m [120 in-lb]

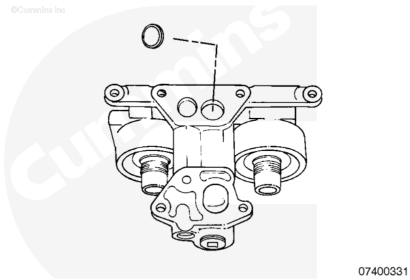
Note : The torque values are the same for the plugs with straight threads and an o-ring seal.

Apply cup plug sealant, Part Number 3375068, or equivalent to the cup plug.

Select the correct size driver.



Install the cup plug.

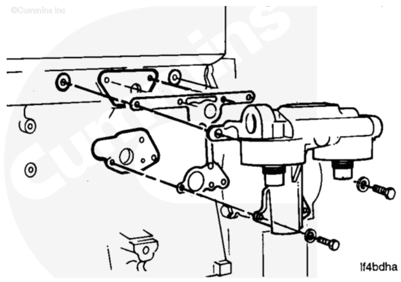


Install

Install the gasket, lubricating oil filter head, and capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

- Install the lubricating oil filters. Refer to Procedure 007-013 Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html).

Last Modified: 22-Oct-2018